Formalizing AI Scientist Systems: A Component Framework and Theoretical Analysis

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Abstract

The emergence of end-to-end systems capable of automating hypothesis generation, experiment execution, and knowledge synthesis represents a significant change in how science is conducted. Despite rapid empirical progress, recent LLM-based *AI Scientist* systems are seldom analyzed within a shared formal vocabulary, which makes principled comparison across architectures and scientific domains difficult. We propose Servo, a six-component framework (S, G, E, V, M, π) comprising the **S**earch space, hypothesis **G**enerator, **E**xperiment executor, **V**alidator, **M**emory, and search **P**olicy (π), grounded in the theory of Partially Observable Markov Decision Processes (POMDPs) and Bayesian Experimental Design (BED). We apply this framework to analyze core AI Scientist systems and their instantiations across seven scientific domains (mathematics, physics, chemistry, materials science, biology, social science, and computer science), and separately examine a complementary artifact infrastructure proposal that addresses the output format and verification layer of such systems. Our analysis surfaces three persistent open problems: (1) the non-automatability of novelty evaluation, which remains dependent on human judgment despite multi-layer automated proxies; (2) the absence of BED-optimal search policies, despite the theoretical availability of tractable approximations; and (3) the underdevelopment of semantic memory M_s, whose absence prevents accumulated experimental experience from improving future search. We formalize the validator V as a reward channel distinct from observation: under tractable, identifying assumptions the discovery loop closes (and validator calibration acts on the reward channel, not the information objective), a validator miscoupled into the belief update drives convergence to the wrong target, and novelty is non-identifiable by any automated validator (Propositions 1–3). Across the surveyed systems and domains, the survey illustrates why trustworthy closed-loop discovery cannot be inferred from mechanical loop closure alone—it turns on which validation channel gates acceptance, whether novelty is automated or human-delegated, and whether memory distills reusable search knowledge—the framework's contribution is this diagnostic separation of bottlenecks that current systems conflate, offered as an organizing distinction rather than a tested association. We read current AI Scientist systems as a progression toward more complete validators.

1 Introduction

AI Scientist systems—agents that independently generate hypotheses, execute experiments, and synthesize knowledge—have proliferated rapidly [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall

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et al., 2025, Feng et al., 2026, Liu et al., 2026], yet the field lacks a shared formal vocabulary. Systems range from tool-augmented synthesis agents (Coscientist [Boiko et al., 2023]) through closed-loop manuscript-writing pipelines (The AI Scientist [Lu et al., 2024, 2026]) to formal-mathematics agents (Aletheia [Feng et al., 2026]), each introducing bespoke terminology that makes systematic cross-system comparison difficult. Without a shared decomposition, questions that bear directly on deployment remain unanswered: *What determines whether a system can operate in a closed loop? Why does autonomous novelty evaluation fail consistently? What is the theoretically optimal search policy?*

We propose Servo[2] (S, G, E, V, M, π), a formal framework grounding AI Scientist systems in POMDP and Bayesian Experimental Design theory (Section 4). We apply it to analyze four core systems, an artifact infrastructure proposal, and instantiations across seven scientific fields (Sections 5, 6); and identify three open problems with falsification criteria (Section 7). The analysis reveals a recurring diagnostic distinction: systems that mechanically close the loop differ sharply in whether their acceptance gate, novelty judgment, and memory structure can support trustworthy discovery claims; we develop the supporting theory in Appendix A. The contribution is thus a diagnostic vocabulary that keeps mechanical closure, validator calibration, novelty judgment, and memory separable rather than collapsing them into a single notion of autonomy.

Scope. We include systems whose primary goal is automated scientific *discovery* through hypothesis-experiment-validation cycles, and exclude specialized predictors without a generation-and-search loop (e.g., AlphaFold [Jumper et al., 2021]). Within that scope we analyze widely-cited representative systems rather than an exhaustive enumeration; this convenience sampling introduces selection bias that we flag in Section 8.

2 Background

Following Kaelbling et al. [1998], we model AI Scientist systems as *partially observable Markov decision processes*: the true state s (unknown structure of nature) is inaccessible; the system maintains a belief $b(s)$ updated by experimental observations. Formally, an AI Scientist POMDP is the tuple $(\mathcal{S}, \mathcal{A}, T, R, \Omega, \mathcal{O}, b_0, \gamma)$, where $\mathcal{A}$ is the action space (hypothesis generation and experiment execution), T is the transition function, R is the reward (scientific value of discovery), $\mathcal{O}$ is the noisy observation process, b_0 is the initial belief, and γ the discount factor; the belief is updated by Bayes' rule, $b'(s') \propto \mathcal{O}(o \mid s', a) \sum_s T(s' \mid s, a)\, b(s)$, and π acts on the belief b rather than the inaccessible state s. Three properties distinguish this from standard control: R cannot be pre-specified (significance is not defined a priori); $\mathcal{S}$ is unbounded; and T is non-stationary (accumulated knowledge changes the exploration frontier). We therefore use the POMDP as an organizing formalism rather than a fully specified model of open-ended science: the formal results of Appendix A are established for the tractable subproblem (Assumption 1) in which these departures are bounded, while the general open-ended case is left open. Bayesian Experimental Design [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] provides the normative criterion for the search policy π: select d^* maximizing Expected Information Gain (EIG),

$$d^* = \arg\max_{d \in \mathcal{D}} \ \mathrm{EIG}(d) = \arg\max_{d} \ \mathbb{E}\left[H(\theta) - H(\theta \mid y, d)\right] \tag{1}$$

where $H(\theta)$ is prior entropy over parameters θ and $H(\theta \mid y, d)$ is expected posterior entropy after observing y under d. EIG can be approximated via variational methods [Foster et al., 2019], amortized policies [Foster et al., 2021], and RL [Blau et al., 2022]. Domain-specific autonomous-science systems already use active-learning or uncertainty-guided approximations—the Robot Scientist's experiment selection [Sparkes et al., 2010] and GNoME's DFT-in-the-loop active learning [Merchant et al., 2023]—but recent open-ended LLM-based systems maintain no explicit posterior over hypotheses or stated EIG objective; that narrower gap is the one we identify in Section 7.

[2] We name the framework after the *servomechanism*: as a servo loop converges on a target by acting on a feedback (error) signal, an AI Scientist closes the discovery loop only when its validator V supplies a reliable feedback signal—the framework's central concern.

3 Related Work

AI Scientist systems. Prior work [Lu et al., 2024, 2026, Schmidgall et al., 2025, Boiko et al., 2023] provides qualitative characterizations of individual systems but no shared framework for cross-system comparison; idea-generation studies [Si et al., 2024, Baek et al., 2024] establish quality baselines without unifying the full discovery loop.

Autonomous-science frameworks. Closed-loop autonomous discovery predates the LLM era: the Robot Scientist [Sparkes et al., 2010] integrated hypothesis generation, robotic experimentation, and statistical validation in yeast functional genomics, instantiating all six Servo components—including an active-learning policy—in physical wet-lab form more than a decade before LLM-based systems. More recent unified frameworks include NovelSeek [Zhang et al., 2025], a closed-loop multi-agent system spanning idea generation to verification across twelve scientific tasks, and Co-Scientist [Gottweis et al., 2026], a multi-agent hypothesis-generation system with experimental biomedical validation. We therefore position Servo not as the first unified framework for automated discovery, but as a formal vocabulary that makes validator completeness, semantic memory, and the POMDP/BED relationship explicit within the recent LLM-agent subfield.

Decision-theoretic foundations. The POMDP framework [Kaelbling et al., 1998] and BED [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] are classical. Bayesian-optimal experiment selection has a long history in automated science—most directly in the Robot Scientist's active-learning policy [Sparkes et al., 2010]—and we build on this tradition, connecting POMDP (structural formulation) to BED (normative policy criterion) for the modern LLM-agent setting rather than claiming priority for the idea itself. The reward specification and specification gaming literatures study the consequences of unreliable reward signals; Servo extends this lens to AI Scientist systems by stratifying V into reliability levels and connecting each level to specific closed-loop failure modes.

Formal models and artifact infrastructure. Taniguchi et al. [2025] propose Collective Predictive Coding (CPC-MS) as a social model of science, recasting scientific activity as decentralized Bayesian inference across a community. CPC-MS and Servo are complementary: CPC-MS provides a *collective* and *social* model in which peer review is a Metropolis-Hastings consensus process; Servo provides the *single-system* and *modular* decomposition that must be instantiated at each node. The three open problems we identify are precisely the necessary conditions for a CPC-MS-compliant collective system to be realizable. Liu et al. [2026] address the artifact layer — output format and verification protocol — rather than the discovery loop itself; we position ARA as a V and M_s infrastructure contribution within Servo, discussed in Sections 5.1 and 7.

4 The Servo Framework

We define an AI Scientist system as a tuple (S, G, E, V, M, π) with the following components:

$$\text{Servo} = (S,\ G,\ E,\ V,\ M,\ \pi) \tag{2}$$

S**: Search Space** The set of candidate hypotheses, programs, molecules, experimental conditions, or other objects subject to exploration. S may be static or dynamically expanding (open-ended).

$G : M \times S \to h$**: Hypothesis Generator** A function that proposes new candidate $h \in S$ conditioned on accumulated memory M and the current state of the search space. G may involve LLM generation, evolutionary mutation, or symbolic search.

$E : h \times P \to o$**: Experiment Executor** A function that executes hypothesis h under protocol P to produce observation o. The fidelity of E ranges from fast computational simulation (low cost, potential surrogate error) to physical wet-lab execution (high cost, ground truth).

$V : o \to r$**: Validator** A function that assigns a reward signal r to observation o. We distinguish four layers of V:

$$
\begin{aligned}
V_{\text{syntax}} &: o \to \{\text{pass}, \text{fail}\} && \text{(deterministic, fast)} \\
V_{\text{semantic}} &: o \to r \in [0,1] && \text{(LLM-based, stochastic)} \\
V_{\text{empirical}} &: o \to r && \text{(reproduction-based)} \\
V_{\text{human}} &: o \to \{\text{novel}, \text{significant}\} && \text{(non-automatizable)}
\end{aligned}
$$

M**: Memory** Structured as (M_e, M_s, M_p). M_e (episodic) stores individual records (h_i, o_i, r_i); M_s (semantic) stores distilled knowledge patterns; M_p (procedural) stores learned protocols. The structure of M directly determines the quality of G and π.

$\pi : b(S) \times M \to h$**: Search Policy** The decision rule for selecting the next hypothesis. The belief state $b(S)$ is updated after each experiment. In the BED interpretation, $\pi^* = \arg\max_d \text{EIG}(d)$ (Equation 1). In practice, current systems use policies ranging from reactive (ReAct) to greedy to tree search.

The operational loop is:

$$
\begin{aligned}
&\pi(b, M) \;\to\; G(M_s, S) \;\to\; E(h, P) \;\to\; V(o) \\
&\qquad\qquad \to\; M \mathrel{+}= (h, o, r) \;\to\; b(S)\text{ update} \;\to\; \text{repeat}
\end{aligned} \tag{3}
$$

We further introduce $H \in [0,1]$ (distinct from the entropy function $H(\theta)$ in Eq. 1) as an auxiliary parameter denoting the degree of human intervention: $H = 0$ for fully autonomous systems and $H = 1$ for systems in which humans assume the role of G or π.

V-completeness as a design dimension. The layers form a partial order under reliability: V_{syntax} deterministic; V_{semantic} stochastic and bias-prone; $V_{\text{empirical}}$ from well-calibrated (DFT) to poorly calibrated (GNN surrogates); V_{human} irreplaceable for novelty but rate-limiting; and V_{formal} (proof kernels, type-checkers) the mechanically-decidable ideal limit, used for mathematics (formal) in Table 2. We define V *completeness* as the highest layer both accessible and reliably calibrated; accessibility without calibration does not count (Agent Laboratory's biased reviewer yields only imperfect completeness). We treat it as a central design variable because it caps what π can learn and what G can be rewarded for: without a reliable V, no policy can distinguish progress from regression. What matters is not that a calibrated layer is *present* but that the layer which *gates acceptance* is calibrated: a calibrated check can be neutralized when a biased semantic or social layer decides acceptance (§8). The cross-domain pattern (Section 6) appears to track V completeness more than G or π sophistication; we develop the relevant formal results next.

The validator as a reward channel. Treating V as distinct from the observation likelihood, we prove three results in a tractable, identifying subproblem (Appendix A): under identifiability and divergent accumulated information the loop closes (Proposition 1); a validator miscoupled into the belief update drives convergence to the wrong target (Proposition 2); and a novelty component invisible to every automated channel is non-identifiable (Proposition 3). A complete validator is thus *necessary* for trustworthy autonomous closure—i.e., for truth-tracking reward assignment—but, by Proposition 1, not sufficient without enough accumulated information; the surveyed systems illustrate these results rather than prove them.

5 Analysis of Core AI Scientist Systems

We apply Servo to four end-to-end systems [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall et al., 2025] and one artifact infrastructure proposal. Table 1 synthesizes the key dimensions; in this small sample, gains in V completeness tend to accompany the transition to closed-loop operation, although the surveyed systems also advance G, E, and π (we revisit this interaction in Section 8).

Across the four systems, validator upgrades or additions accompany the move to closed loops; this does not mean G or π advances are absent—several systems also upgrade ideation, tool use, and search—but in these cases validator reliability is the constraint that gates whether such advances can be exploited in a closed loop. Coscientist exhibits only constrained task-level feedback (measured outcomes inform later choices in reaction-optimization experiments [Boiko et al., 2023]); its discovery loop is therefore *partial* rather than absent, with validation of novelty and significance remaining

	Robot Sci.	Coscientist	AI Sci. (2024)	AI Sci. (2026)	AgentLab	NovelSeek
Novelty measure	Stat. test	None	LLM self-score	3-layer	Human-deleg.	Perf. gain
Loop closed?	Yes (wet-lab)	Partial	Yes (greedy)	Yes (tree)	Partial	Yes (comp.)
π type	Active learn.	ReAct	Greedy	Tree search	Human/seq.	Feedback
V completeness	V_e	$V_h + V_e$	V_s (biased)	$V_e + V_s + V_h$	V_s (biased)	V_e
M structure	$M_e + M_s$	Ephemeral	M_e	$M_e + M_s$	M_e	M_e
H (approx.)	Low	G ($\approx$1)	Low ($\approx$0)	Low ($\approx$0)	G+π ($\approx$0.8)	Low

Table 1: Comparative Servo analysis of core AI Scientist systems, including the pre-LLM Robot Scientist [Sparkes et al., 2010] and the recent unified framework NovelSeek [Zhang et al., 2025]. V_s=semantic, V_e=empirical, V_h=human.

human-mediated. The AI Scientist (Sakana) adds V_s via a 5-ensemble GPT-4o reviewer (balanced accuracy 0.65 vs. human 0.66 [Lu et al., 2024]), enabling the first *structurally* closed loop despite greedy π, though on a biased, uncalibrated V. The AI Scientist (Nature) stacks V_e (replication nodes with mean $\pm$ s.d.), V_s (automated reviewer, balanced accuracy 0.69 [Lu et al., 2026]), and actual peer review (V_h via ICLR workshop), achieving the most complete V in the class. Agent Laboratory delegates V to V_s (NeurIPS-guideline LLM reviewer) but documents $+2.3$-point systematic over-estimation relative to human PhD students—illustrating that adding a V layer does not guarantee V completeness if the layer is unreliable. Within this small sample we observed no case where G or π advances alone improved loop feasibility without a corresponding V improvement, though a four-system sample cannot establish this as a general impossibility.

5.1 Artifact Infrastructure: Agent-Native Research Artifacts

Liu et al. [2026] address a complementary problem—what the *output* of the loop should be and how to verify it. Agent-Native Research Artifacts (ARA) replace the narrative paper with a four-layer ontology (`/logic`, `/src`, `/trace`, `/evidence`); the ARA Seal automates V_{syntax}, V_{semantic} (Rigor Auditor, 82.6% detection), and $V_{\text{empirical}}$ (sandboxed reproduction), reserving V_{human} for significance and novelty [Liu et al., 2026]. Motivating figures: only 45.4% of papers fully specify reproduction requirements and 90.2% of extension cost is failed exploration. The `/trace` layer externalizes M_e/M_s, but preserved traces both accelerated and anchored downstream agents—richer M_e does not monotonically improve π.

6 Domain Analysis

Table 2 is an illustrative domain map: it uses the Servo vocabulary to expose how validation bottlenecks differ across seven scientific fields, the bottleneck following from each field's V type (Propositions 1–3). To illustrate scale: Aletheia [Feng et al., 2026] attempted 700 Erdős problems; of the 200 solutions selected for human audit, 13 (6.5%) were rated meaningfully correct; GNoME [Merchant et al., 2023] predicted 2.2 million structures (380,000 stable) using DFT-in-the-loop active learning. Systems analyzed span mathematics (formal: [Romera-Paredes et al., 2024, Xin et al., 2024]; NL: [Feng et al., 2026]), physics [Udrescu and Tegmark, 2020, Cranmer et al., 2020], chemistry [Bran et al., 2024, Sprueill et al., 2024], materials [Merchant et al., 2023], biology [O'Donoghue et al., 2023], social science [Manning et al., 2024, Aher et al., 2023, Park et al., 2023], and CS/algorithms [Real et al., 2020, Sartori and Blum, 2024, Jaber and Jaber, 2026].

The pattern is largely shaped by V type. Formal mathematics has a binary V_{formal} that makes RL stable and the loop fully closed; natural-language proof reintroduces V_{human} (Aletheia: 6.5% audited-correct). Chemistry and biology close the computational loop but leave the physical loop open at the measurement boundary, where characterization stays human-delegated. Materials science is the one domain whose π approximates BED, since DFT is a well-calibrated oracle. Social science suffers circular validity when G and E share a model [Aher et al., 2023], and CS/algorithms close the loop via reliable V_{emp} but lack interpretability.

7 Open Problems

Our analysis identifies three open problems that persist across all systems and domains. For each we state a *proposed* falsification test—a concrete design whose outcome would refute the correspond-

Domain	V type	V reliability	Loop	π	Primary bottleneck
Math (formal)	V_{formal}	Perfect oracle	Fully closed	RL/MCTS	Pretraining contamination
Math (NL)	$V_{\text{sem}} + V_h$	Imperfect	Closed (low quality)	Seq. refine	Hallucination; plagiarism
Physics	$V_{\text{emp}} + V_h$	Medium	None/partial	Deterministic	No cross-problem M
Chemistry	V_{emp} (surr.)	Medium	Comp. only	Beam	Wet-lab loop open
Materials	V_{emp} (DFT)	High	Comp. closed	Uncert. samp.	DFT wall-time; synth. lag
Biology	$V_{\text{emp}} + V_h$	Low–med	Partial	None	V metrics insufficient
Social Sci.	Stat. test	Low–med	Partial	Factorial	Circular validity
CS / Algo.	V_{emp}	High	Fully closed	Evol./LLM	Interpretability absent

Table 2: Illustrative Servo domain mapping across seven fields (per-domain coding in `domain_systems.csv`). V reliability refers to agreement with ground truth.

ing claim—while noting that none are executed here; they specify what evidence would settle the question rather than constituting completed experiments.

7.1 The Non-Automatability of Novelty

Novelty cannot be reliably evaluated by automated means: every system from Coscientist (no mechanism) to The AI Scientist Nature (three-layer validation) converges on this conclusion. Two failure modes are documented: LLM self-evaluation bias (systematic over-estimation; hallucinated results) and pretraining contamination (systems may "discover" results implicit in training data, as Aletheia documents for Erdős problems). The ARA Seal [Liu et al., 2026] demarcates the automated frontier: its Rigor Auditor achieves 82.6% average detection across five structural violation types, but detects only 22% of orphaned-experiment violations specifically (the two rates measure different denominators and are not complements). It cannot cross the novelty boundary. Concretely: Agent Laboratory's LLM reviewer over-estimated paper quality by $+2.3$ points on the NeurIPS review scale relative to human PhD students; The AI Scientist (Sakana) hallucinated ablation tables; and of the 200 Erdős solutions selected for human audit (from 700 attempted), only 13 (6.5%) were rated meaningfully correct despite surfacing many superficially plausible candidates.

Proposed falsification test (design; not yet executed): The thesis is refuted if an automated validator matches a panel of ≥ 3 domain experts on novelty discrimination—Cohen's $\kappa \geq 0.80$ against the expert majority—on a benchmark of published-versus-withheld findings whose training-set membership is fixed by a pretraining cutoff date, across ≥ 3 distinct domains. Constructing the contamination-controlled benchmark and the expert baseline is itself an open task; we specify the instrument and threshold but do not run the test.

Novelty, significance, and correctness are distinct properties that current systems collapse into one scalar (e.g., The AI Scientist's 1–10 score [Lu et al., 2024]); a viable automated novelty V would need a contamination-controlled reference corpus, architectural separation between G and V, and a definition of "novel to the field" beyond lexical training-set membership—none of which current systems provide.

7.2 The BED–Practice Gap for π

Equation 1 specifies the theoretically optimal policy, yet no surveyed AI Scientist system implements it: all deployed policies in the systems analyzed here are heuristic (ReAct, greedy, tree search, human-guided). The partial exception is GNoME [Merchant et al., 2023], whose deep-ensemble uncertainty sampling approximates EIG over a tractable GNN posterior—but only because DFT provides a well-calibrated V and the hypothesis space admits a learnable probabilistic model, conditions absent from the LLM-based AI Scientist systems surveyed here, where the "posterior" is implicit in model weights [Foster et al., 2021]. Recent work (BED-LLM [Choudhury et al., 2026]) derives a usable EIG posterior from an LLM's predictive distributions, but for conversational queries rather than scientific experiment design—so the gap is integration with reliable E and V, not impossibility. Concretely, every system in Table 1 degrades to heuristic π: Coscientist selects the next tool reactively, The AI Scientist (Sakana) greedily avoids archive similarity, and Agent Laboratory follows a fixed sequential schedule—none improve with additional experimental data, the failure mode EIG-optimal π would avoid.

Proposed falsification test (design; not yet executed): The gap is refuted if a system explicitly computes and maximizes EIG over a posterior $p(\theta \mid y)$ whose hypothesis space admits no closed-form or finite-ensemble surrogate—operationalized as a posterior defined directly over natural-language hypotheses or program text rather than over a hand-specified low-dimensional model—thereby excluding tractable-model proxies such as GNoME's deep-ensemble policy. We give this boundary as the measurable criterion but do not test it.

Architecturally, EIG needs a posterior $p(\theta \mid y)$, but in LLM systems θ is implicit in frozen weights with no per-experiment update. Either one maintains an explicit probabilistic surrogate (GNoME's route, viable only where a calibrated input–outcome model exists) or uses amortized EIG estimation [Foster et al., 2021], which remains unsolved for open-ended $\mathcal{S}$.

7.3 The Underdevelopment of Semantic Memory M_s

Current systems store raw experiment records (M_e) but do not distill them into transferable patterns (M_s): a system with 1,000 experiments does not search more intelligently than one with 10. The ARA `/trace` layer [Liu et al., 2026] is the closest approximation, but traces accelerated extension performance while also anchoring capable agents to prior solution paths—M_s structure must be paired with selective forgetting mechanisms to be beneficial. Concretely, GNoME's GNN—one of the few systems where $G(M_s, S)$ genuinely improves—accumulates domain-specific thermodynamic knowledge after 460K DFT calculations, yet this knowledge does not transfer to chemistry or biology; the system is no more capable in any domain it has not explicitly trained on.

Proposed falsification test (design; not yet executed): The bottleneck is refuted if, with the base model held fixed, a system's G-generated hypothesis quality improves monotonically with the number of stored prior experiments on a held-out task family it was not trained or built for—measured as expert-rated quality or downstream task success against a pre-registered effect-size threshold, with an ablation that removes M_s. We specify the protocol but do not execute it.

The hard M_s problems arise when memory is lossy, compressed, conflicting, or strategically filtered—each of which can degrade G rather than improve it. GNoME shows within-domain accumulation works, but its knowledge does not transfer across domains; building M_s that generalizes, forgets selectively, and tolerates conflicting evidence is the open precondition for systems whose $G(M_s, S)$ improves with scale.

The three open problems are not independent. A system with complete V and tractable M_s would have the prerequisites for an EIG-optimal π: complete V supplies the calibrated reward signal that reward-guided policy selection needs, and tractable M_s supplies the hypothesis-space model over which information gain can be estimated. Whether they suffice in non-stationary or unbounded S is open, but they are necessary: the problems converge on a single structural gap where progress on one unlocks the others. They are also the necessary conditions for a collective AI co-scientist [Taniguchi et al., 2025]: $V_{\text{collective}}$ needs solved novelty automatability, a principled collective π a closed BED–practice gap, and shared memory w a developed M_s—with ARA [Liu et al., 2026] a plausible entry point that externalizes M_s and calibrates the sub-novelty layers of V.

8 Discussion

Validator design as a constraint. Our analysis suggests V design is a necessary constraint on closed-loop operation: unreliable validation prevents a system from learning whether its outputs represent progress. The surveyed systems do not isolate V from G, E, M, and π, so we read V completeness as a recurring bottleneck, not an established causal priority. The practical implication still stands: before optimizing G or π, designers should ask what level of V completeness the domain actually supports.

Human oversight. Rather than treating $H \in [0, 1]$ as a quantity to minimize, our analysis suggests explicit human integration at the V_{human} and π levels is both practically necessary and theoretically appropriate: human judgment accesses an information source no automated proxy yet replicates, particularly for the non-automatizable components of novelty and significance. The goal is not full autonomy but *appropriate* autonomy—closing the loop where V is reliable while preserving human judgment where it is not.

Towards collective AI co-scientists. All systems analyzed are single-agent loops; extending Servo to a multi-agent collective with shared memory w and a collective validator [Taniguchi et al., 2025] requires exactly the three open problems above to be solved, with ARA [Liu et al., 2026] a plausible entry point for w.

Limitations. Servo abstracts away several factors: the cost structure of E (orders of magnitude across domains), the non-stationarity of S, and the single-agent framing's social layer [Taniguchi et al., 2025]. The cross-domain comparison also relies on publicly reported descriptions that may not capture all implementation details. More fundamentally, the V-completeness pattern is a cross-system *association*, not an established causal determinant, and the surveyed systems are a convenience sample subject to selection and survivorship bias. Mechanical loop closure moreover does not require completeness (AI Scientist v1 and NovelSeek close a computational loop at low completeness), so the claim concerns *trustworthy* closure, which we do not measure independently of V; we present it as an organizing hypothesis, not a tested association, and the supplement adds a descriptive check coding `trustworthy_closure` from signals external to the validator (direction-consistent but, at six systems, far too small for inference). To make the coding auditable we release the rubric, coding sheets with a representative source quote and citation per row (core systems and the cross-domain Table 2), and validator scripts that reject any uncited or unquoted row. Three independent, model-pinned, blind LLM-based coder agents from different vendors (Anthropic, OpenAI, Google)[3] coded 14 systems; inter-coder agreement (Fleiss' κ over the 13 jointly-coded systems) is substantial-to-perfect on most constructs (calibration κ=0.79, loop status κ=0.74, semantic-validation presence κ=1.0; mean κ=0.68) and only fair on the holistic V-completeness ordinal (κ=0.39, on which independent coders also disagree with the author), reported as an automated multi-rater transparency measure rather than human-level reliability. A counterexample search found no system that is V-incomplete yet achieves *trustworthy* closed-loop discovery: the lone structural near-miss (AI Scientist v1) closes its loop on an admittedly biased validator but yields unreliable output. The conceptual boundaries are thus made precise, under stated assumptions, by Propositions 1–3; we use the survey to expose failure modes in validator *architecture*, not to estimate an association. Blind coders split from the author labels (themselves AI-coded, human-directed) on whether AI Scientist (Nature 2026) has a "calibrated" validator—the old label conflated a calibrated layer being *present* with the *gating* layer being calibrated. Re-coding the split over all 14 systems (two vendors; supplement) resolves it (author and both coders agree the layer is present but non-gating); gating calibration is more clearly specified by this two-vendor recoding (raw agreement 13/14, Cohen's κ=0.86; an automated robustness check, not expert-validated reliability), clarifying why mere calibrated-layer presence is insufficient, and no system has a calibrated *automated* novelty gate, grounding the novelty open problem.

9 Conclusion

We introduced Servo (S, G, E, V, M, π), a formal framework connecting AI Scientist systems to POMDP and BED theory, applied to four core systems, an artifact infrastructure proposal, and seven domains. The central pattern in our AI-coded, human-directed survey is that the calibration of the loop's *decisive* validation channel is what most cleanly distinguishes trustworthy from merely mechanical closure—an observational distinction, not a proven causal hierarchy. Propositions 1–3 support only narrower claims (closure needs an identifying observation model and sufficient information; calibration governs the reward/policy channel, not posterior consistency; novelty is unidentified when no automated channel carries it), so the formalism separates observation, reward validation, and human novelty judgment rather than proving V completeness is the primary determinant.

These bottlenecks are structural—missing validator and memory architecture—and Servo names the vocabulary for what closing them requires.

[3] Coders: Claude Code (`claude-opus-4-8`, Anthropic), Codex CLI (`gpt-5.5`, OpenAI), and Antigravity (`gemini-3.1-pro`, Google), each run headless and model-pinned; exact invocations and per-model outputs are released in the supplement.

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Conference For AI Scientists 2026 - AI Involvement Checklist

This checklist documents the role of AI systems at each stage of this research. The checklist has two parts: a **Research Stage Assessment** (items 1–4) rating AI involvement and iteration effort, and an **AI System Documentation** section (items 5–6).

Research Stage Assessment

1. **Hypothesis development.**
   Answer: **[B]**
   Iteration Effort: **[Medium]**
   Explanation: The core Servo framework tuple (S, G, E, V, M, π), the V-completeness thesis, the H parameter, and the three open problems were conceived by the human author. A frontier LLM agent elaborated framework components, identified literature-level instantiations, and proposed theoretical connections. The human author directed all conceptual decisions and made several substantive corrections to the AI's elaborations.
2. **Experimental design and implementation.**
   Answer: **[B]**
   Iteration Effort: **[Medium]**
   Explanation: This submission contains no computational experiments. The methodology for cross-system analysis (system selection criteria, component extraction protocol, domain comparison structure) was designed by the human author. A frontier LLM agent executed literature retrieval, extracted framework components from primary sources, and performed citation-level fact verification across four verification sweeps.
3. **Analysis of data and interpretation of results.**
   Answer: **[C]**
   Iteration Effort: **[Medium]**
   Explanation: The LLM agent extracted Servo components for all analyzed systems from primary sources and performed citation-level fact verification (4 verification sweeps). The human author reviewed all extractions, corrected mischaracterizations, and synthesized the cross-domain pattern. The V-completeness thesis emerged from the human author's synthesis.
4. **Writing.**
   Answer: **[C]**
   Iteration Effort: **[High]**
   Explanation: The LLM agent drafted the majority of the manuscript text through multiple revision cycles; the human author provided key conceptual passages (framework definitions in §4, core open problems in §7), directed structural decisions, reviewed all drafts, and made final editorial judgments.

AI System Documentation

5. **AI systems used.**
   Description: A frontier large language model with code execution and long-document processing capabilities was used as the primary research and writing assistant. The system operated as an interactive CLI agent with file-system access, able to read primary sources, execute and debug Python simulations, and iteratively revise LaTeX manuscripts.
6. **Observed AI Limitations.**
   Description: (1) Systematic citation fabrication: the agent generated plausible but non-existent quotes and statistics, requiring four dedicated citation-verification sweeps using parallel sub-agents. (2) Analytical overconfidence: the agent initially overclaimed empirical significance for mechanically-entailed theoretical consequences; multiple adversarial review rounds were needed to correct framing and remove overclaims. (3) Context drift: over long sessions, the agent occasionally regressed previously corrected errors, requiring explicit re-verification of integrity items before each commit.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction explicitly state the three contributions (framework, cross-domain analysis, open problems). The propositions under explicit assumptions (§A) clarify the separation of observation, reward validation, and novelty judgment rather than establishing a V-completeness thesis, and the cross-domain survey is presented as illustration; the paper delimits both the assumptions and the survey's illustrative role in §A and §8.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Limitations are discussed in §8 (framework abstracts domain cost structures, social layer, non-stationarity) and throughout the domain analysis (§6).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

   Answer: [Yes]

   Justification: Appendix A states three propositions, each with its assumptions (Assumption 1, Definition 1) and a proof sketch grounded in standard results on Bayesian experimental design and posterior consistency under model misspecification. The cross-domain survey is presented as illustration of these results, not as independent empirical proof.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: This paper contains no computational experiments. The cross-domain analysis approach (system selection and component extraction from primary sources) is described in §5 and §6; machine-readable extraction sheets (core systems and the seven-domain Table 2) with a representative source quote and citation per row and citation-validating regeneration scripts, explicit inclusion/exclusion criteria, multi-vendor blind coding over 14 systems with Fleiss' κ (plus a two-vendor Cohen's κ and recode sensitivity for the calibration re-coding), and a counterexample search are released as supplementary material (see §8). All analyzed papers are publicly accessible.

5. **Open access to data and code**

   Question: Does the paper provide open access to data and code?

   Answer: [No]

   Justification: This submission contains no computational experiments. All analyzed papers are publicly accessible, and the framework can be applied following the component definitions in §4.

6. **Experimental setting/details**

   Question: Does the paper specify all details necessary to understand the results?

   Answer: [Yes]

   Justification: The cross-domain analysis methodology is described in §5 and §6. No computational experiments were performed in this submission.

7. **Experiment statistical significance**

Question: Does the paper report error bars or statistical significance information?

Answer: [NA]

Justification: This paper contains no computational experiments and reports no statistical results requiring error bars or significance tests.

8. **Experiments compute resources**

   Question: Does the paper provide sufficient information on compute resources?

   Answer: [NA]

   Justification: This submission contains no computational experiments. The meta-analysis requires only standard document processing and no specialized compute.

9. **Research Ethics**

   Question: Does the research conform with the NeurIPS Code of Ethics, except where CAISc's AI authorship policies apply?

   Answer: [Yes]

   Justification: This is a meta-analysis of published systems with no human subjects, no proprietary data, and no safety risks. AI authorship is documented per CAISc policy.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive and negative societal impacts?

    Answer: [Yes]

    Justification: §8 addresses both the potential for accelerating scientific discovery and the risk of premature convergence under widespread AI Scientist deployment (particularly via the M_s underdevelopment problem and novelty non-automatability).

A Formal Analysis of Validator Completeness

We make the role of V precise by separating two objects that are easily conflated: the *observation likelihood* $p(o \mid \theta, d)$, which drives the Bayesian belief update, and the *validator* $V(o)$, a reward estimate that the policy π uses to score outcomes. The results hold in a tractable subproblem; the open-ended regime is left open.

Assumption 1 (Tractable, identifying subproblem)**.** *(i) the search space S is bounded and the parameter space Θ is compact; (ii) T is stationary; (iii) the prior $p(\theta)$ has support containing the true $\theta^\star$; (iv) the observation model $p(o \mid \theta, d)$ is well-specified and identifies θ (for $\theta \neq \theta'$ some reachable design separates them); and (v) the realized design sequence is* uniformly informative*—infinitely often it selects a reachable design whose Kullback–Leibler separation between any two surviving hypotheses is bounded below by a fixed $\delta > 0$, equivalently the accumulated information $\sum_t \mathrm{EIG}(d_t)$ diverges.*

Definition 1 (Validator as reward channel)**.** *$V(o)$ estimates the true reward $R(s, a)$; it is* calibrated *if $\mathbb{E}[V(o) \mid s, a] = R(s, a)$ and* biased *if $\delta(s, a) = \mathbb{E}[V(o) \mid s, a] - R(s, a) \neq 0$ for some (s, a). V is distinct from the update likelihood $p(o \mid \theta, d)$ unless explicitly coupled into it.*

Proposition 1 (Sufficiency for closure)**.** *Under Assumption 1 the posterior concentrates on $\theta^\star$ (the loop closes), and the one-step EIG rule (Equation 1) is myopically optimal for the information objective—both independently of the validator, since V is a reward channel while EIG is a functional of the prior and likelihood alone.*

Proof sketch. A well-specified, identifying likelihood with *divergent* accumulated information on a compact Θ (Assumption 1(i),(iv),(v)) is the standard sufficient condition for Bayesian posterior consistency [Ghosal and van der Vaart, 2017], giving $b_t \to \delta_{\theta^\star}$. Calibration is required for neither consistency nor EIG-optimality (both depend only on the prior and likelihood); it concerns the separate reward channel (we make this precise below). The uniform-information clause (v) is essential: a sequence with $\mathrm{EIG}(d_t) > 0$ but $\sum_t \mathrm{EIG}(d_t) < \infty$ leaves the posterior short of a point mass even under identifiability and calibration. □

Proposition 2 (Drift under misspecified coupling)**.** *Suppose the validator enters the belief update, i.e. the effective update is $q_\theta(o) \propto p(o \mid \theta, d)\, \exp(\beta V(o))$ with coupling $\beta > 0$. If V depends on* *the outcome in a θ-dependent (non-affine) way, the posterior concentrates on the Kullback–Leibler projection $\theta^\dagger \neq \theta^\star$ [Berk, 1966]: the loop closes computationally but on the wrong target. A θ-independent affine transformation of V (e.g. a constant additive bias) cancels in the log-partition and leaves $\theta^\star$ unchanged; and when V is a pure policy channel ($\beta = 0$) no validator transformation moves the target—it can at most degrade design choice and slow information acquisition (interacting with Assumption 1(v)).*

Proof sketch. A coupled, misspecified update likelihood places the posterior at its KL minimizer [Berk, 1966]; the log-partition $\log \int p(o \mid \theta, d)\, e^{\beta V(o)}\, do$ relocates that minimizer iff its dependence on θ changes, which a θ-independent affine term does not. With $\beta = 0$ the update likelihood is unchanged, so $\theta^\star$ is invariant to V. □

Proposition 3 (Non-identifiability of novelty)**.** *Let θ_{nov} be a novelty/significance component that is a priori independent of the identified components. If the channel likelihood of every automated validator is independent of θ_{nov}, then no automated validator admits a consistent estimator of θ_{nov}, and loop closure on novelty requires V_{human}.*

Proof sketch. If the likelihood does not depend on θ_{nov} and the prior does not couple it to identified components, the posterior marginal on θ_{nov} equals the prior for all data; θ_{nov} is unidentified and admits no consistent estimator. □

These results bound the validator-completeness reading rather than overstate it. A complete validator is *necessary* for autonomous closure—it supplies the calibrated reward and, for $V_{\text{empirical}}$, often the identifying, well-specified channel Assumption 1 requires—but Proposition 1 shows convergence *also* needs sufficient accumulated information, Proposition 2 shows that drift requires an incorrectly *coupled* validator (not a mere ranking shift), and Proposition 3 isolates the one component no automated validator can resolve. Finally, information-optimal and reward-optimal design must be distinguished: the EIG-maximizing and expected-reward-maximizing designs coincide only when expected validator reward preserves the EIG ordering, and diverge even for a perfectly calibrated V (e.g. $\text{EIG}(d_1) = 1 > \text{EIG}(d_2) = 0.1$ but $\mathbb{E}[V(O_{d_1})] = 0 < \mathbb{E}[V(O_{d_2})] = 10$); calibration governs the reward/policy channel, not which design maximizes EIG.